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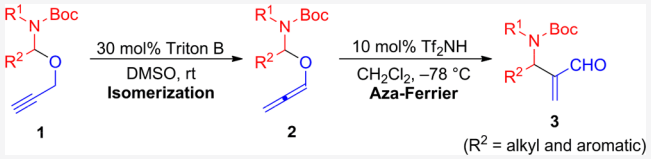
Brønsted Acid-Catalyzed Aza-Ferrier Reaction of *N,O*-Allenyl Acetals: Synthesis of β-Amino-α-methylene Aldehydes

Eiji Tayama* and Yuki Ishikawa

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ABSTRACT: A Tf_2NH-catalyzed aza-Ferrier reaction of *N,O*-allenyl acetals was reported. This protocol provided various types of β-amino-α-methylene aldehydes as the products. The *N,O*-allenyl acetal substrates were easily prepared by base-induced isomerization of *N,O*-propargyl acetals with Triton B. The *N,O*-propargyl acetals were prepared from the corresponding aldehydes or lactams. Further synthetic applications of the products were also described.

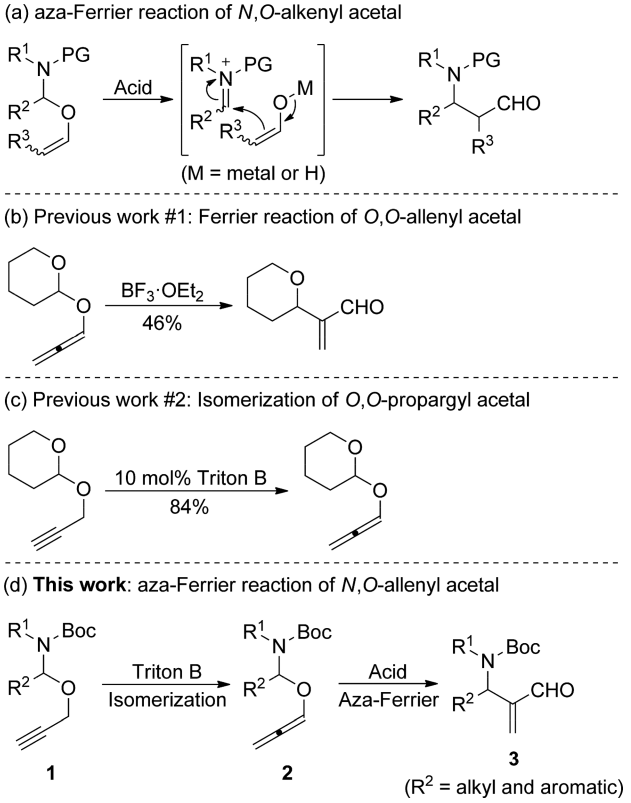


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The acid-catalyzed aza-Ferrier reaction of *N,O*-alkenyl acetals is an atom-economical reaction that proceeds via acid-catalyzed cleavage of the C−O bond to generate the corresponding iminium ion as an electrophile and the enolate (Lewis acid) or the enol (Brønsted acid) as a nucleophile (Scheme 1a).[1] Sequential C−C bond formation provides the corresponding β-amino aldehydes, which act as valuable building blocks for the synthesis of biologically active compounds such as β-amino acids and alkaloids. Initial studies on this reaction were carried out by Frauenrath et al.[2] Their group reported the Lewis acid-catalyzed reaction of *N,O*-alkenyl acetals to afford α-substituted-β-amino aldehydes. Our group applied this protocol to the reaction of *N,O*-1,3-dienyl acetals as substrates and obtained β-amino-β′,γ′-unsaturated aldehydes.[3] The enantio- and diastereoselective reaction was achieved by Terada's group using a chiral Brønsted acid catalyst.[4] However, from a synthetic point of view, this reaction has a disadvantage in the preparation of *N,O*-alkenyl moieties because efficient preparative methods of these moieties have been limited. The preparations were conducted using *N,O*-allylic acetals as a precursor by (i) base-induced isomerization, (ii) transition-metal-catalyzed isomerization, or (iii) base-promoted elimination. These methods limit the substrate and product scopes; therefore, further studies and development of the aza-Ferrier reaction have not been advanced.

To progress the study on the aza-Ferrier reaction, we focused on two previous works: (i) Lewis acid-catalyzed Ferrier reaction of *O,O*-allenyl acetals into β-alkoxy-α-methylene aldehydes reported by Ricci et al. (Scheme 1b)[5] and (ii) efficient and simple isomerization of *O,O*-propargyl acetals into *O,O*-allenyl acetals with benzyltrimethylammonium hydroxide (Triton B) reported by Ishikawa et al. (Scheme 1c).[6] These works gave us the idea of an aza version of these transformations from *N,O*-propargyl acetals **1** into β-amino-α-methylene aldehydes **3** via *N,O*-allenyl acetals **2** (Scheme 1d).[7] Although the structures of products **3** are in the same category as those of the products obtained by the aza-

Scheme 1. Flow of Ideas: Aza-Ferrier Reaction of *N,O*-Alkenyl Acetals to *N,O*-Allenyl Acetals



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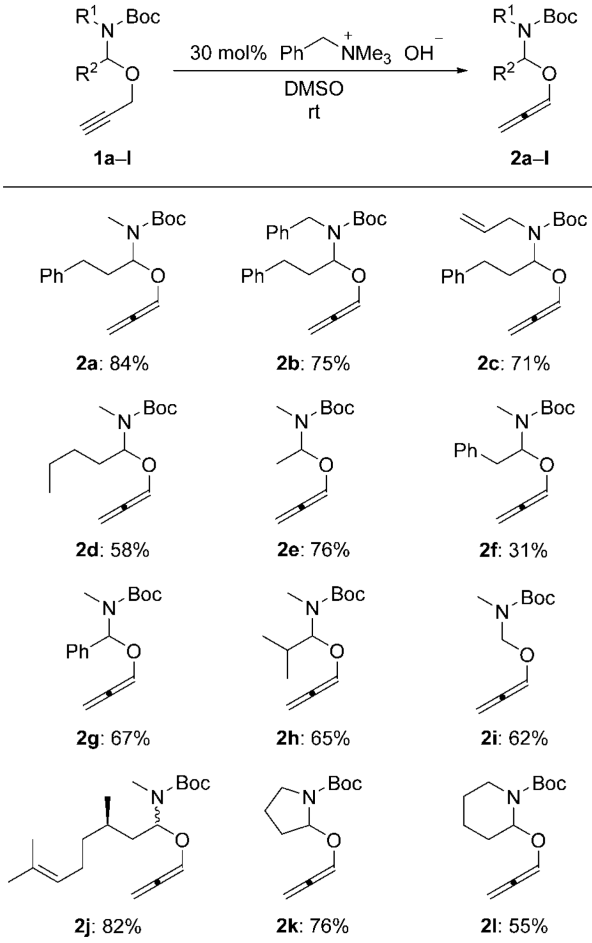
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Morita–Baylis–Hillman (aza-MBH) reaction,[8] we believed that our protocol would expand the product library. The main product examples of aza-MBH reactions are limited to β-aryl derivatives because the reaction requires the use of an imine as the substrate. Preparable and chemically stable aromatic imines are strongly preferred. Therefore, we started to investigate the acid-catalyzed aza-Ferrier reaction of *N*,*O*-allynyl acetals **2** to provide various types of β-amino-α-methylene aldehydes **3**.

First, we tried to prepare *N*,*O*-allenyl acetal **2a** by the traditional base-induced isomerization of *N*,*O*-propargyl acetal **1a**[9] with *t*BuOK.[10] The exact reason is unclear currrently, and the yield of desired **2a** was low, with the recovery of **1a** and small amounts of decomposition materials. Thus, we attempted to use Triton B for the isomerization according to the literature,[6] as depicted in Scheme 1c. This reagent was found to be quite effective for isomerization of **1a** (Scheme 2).

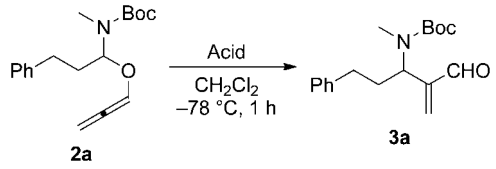
Scheme 2. Preparation of *N*,*O*-Allenyl Acetals 2



A commercially available MeOH solution of Triton B was evaporated and dried under a vacuum. The residue and **1a** in DMSO were stirred at room temperature to give desired **2a** in an 84% yield. Other derivatives **2b–l** were also prepared under the same conditions.

With the substrates in hand, we investigated the aza-Ferrier reactions of *N*,*O*-allenyl acetal **2a** in CH_2Cl_2 at −78 °C for 1 h in the presence of 10 mol % Lewis or Brønsted acid catalysts (Table 1). First, we selected the standard Lewis acid $BF_3{\cdot}OEt_2$, and the reaction gave desired β-amino-α-methylene aldehyde **3a** in a 53% yield (entry 1). The use of $Sc(OTf)_3$ did not improve the yield of **3a** (entry 2). $Cu(OTf)_2$ did not show catalytic activity under the conditions used (entry 3). The reactions in the presence of standard Brønsted acids such as *p*-TsOH·H_2O and PPTS resulted in failure (entries 4 and 5). Decomposition or recovery of substrate **2a** was observed. However, when the reaction was carried out with the stronger Brønsted acid Tf_2NH,[11] the yield of **3a** was dramatically improved to 93% (entry 6). These conditions were found to be applicable for an experiment on a scale greater than 1 mmol (entry 7, 93% yield). With lower catalytic amounts (1 or 5 mol %) of Tf_2NH, the yields of **3a** were decreased (entries 8 and 9).

Table 1. Screening of Acid Catalysts[a]



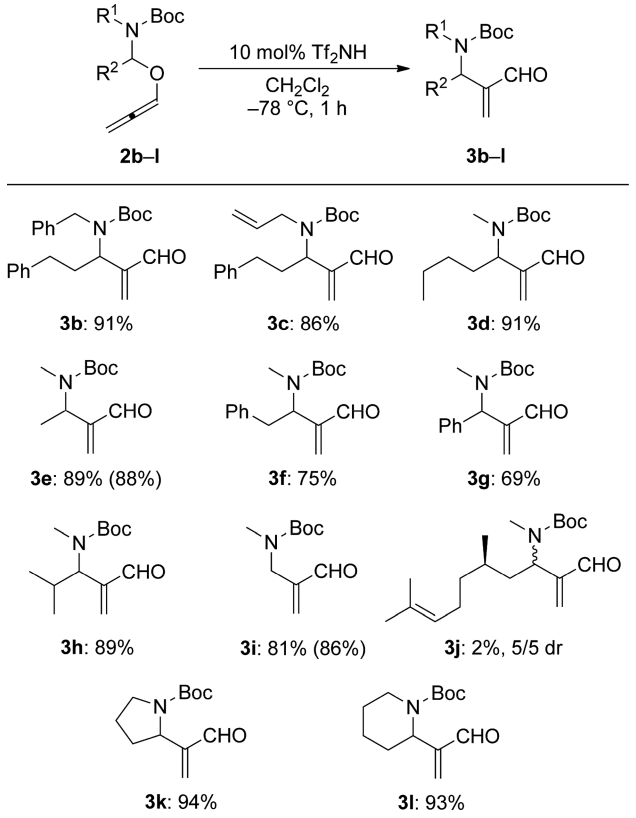
entry	acid (mol %)	yield (%)[b]
1	$BF_3{\cdot}OEt_2$ (10)	53
2	$Sc(OTf)_3$ (10)	32
3	$Cu(OTf)_2$ (10)	0[c]
4	*p*-TsOH·H_2O (10)	6[d]
5	PPTS (10)	0[e]
6	Tf_2NH (10)	93
7[f]	Tf_2NH (10)	93
8	Tf_2NH (5)	83
9	Tf_2NH (1)	70

[a]Unless otherwise noted, the reactions were performed using approximately 0.3 mmol of **2a**. [b]Isolated yield. [c]95% recovery of **2a**. [d]3-Phenylpropionaldehyde was obtained in a 19% yield. [e]88% recovery of **2a**. [f]1.25 mmol of **2a** was used.

With the optimized conditions in hand, we prepared a series of substrates **2b–l** and carried out their reactions to define the scope and limitations of the present aza-Ferrier reaction of *N*,*O*-allenyl acetals (Scheme 3). The reactions of substrates **2b** and **2c** bearing removable substituent R^1 on the nitrogen atom, such as *N*-benzyl and *N*-allyl, proceeded to give **3b** and **3c**, respectively, in good yields. Next, we investigated the reactions of **2d–j** bearing different types of substituent R^2 to obtain various types of β-amino-α-methylene aldehydes **3d–j**. The reactions of substrate **2** with a simple substituent, such as **2d** (R^2 = *n*Bu) and **2e** (R^2 = Me), proceeded with excellent yields (**3d**, 91%; **3e**, 89%). Interestingly, the reaction of **2f** (R^2 = CH_2Ph) produced the corresponding product **3f** in 75% yield, although the intermediate iminium ion generated from **2f** has acidic and isomerizable benzylic protons. To compare with the aza-MBH reaction, we investigated the reaction of aryl derivative **2g** (R^2 = Ph) to produce **3g**, which is the standard structure of the aza-MBH reaction product. The reaction rate observed by TLC analysis was slow; thus, the yield of **3g** was low (69%). The reactions were not affected by a secondary alkyl substituent, such as isopropyl **2h** (R^2 = *i*Pr) and unsubstituted **2i** (R^2 = H) derivatives, and produced **3h** (89%) and **3i** (81%), respectively. We considered the reaction of the *tert*-butyl derivative (R^2 = *t*Bu); however, the preparation of the corresponding substrate **2** was unsuccessful. A failure example was the reaction of (+)-citronellyl derivative **2j** (R^2 = (+)-citronellyl). The desired **3j** was obtained in only a 2% yield, with the formation of various side products observed

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Scheme 3. Scope of the Reaction[a,b]



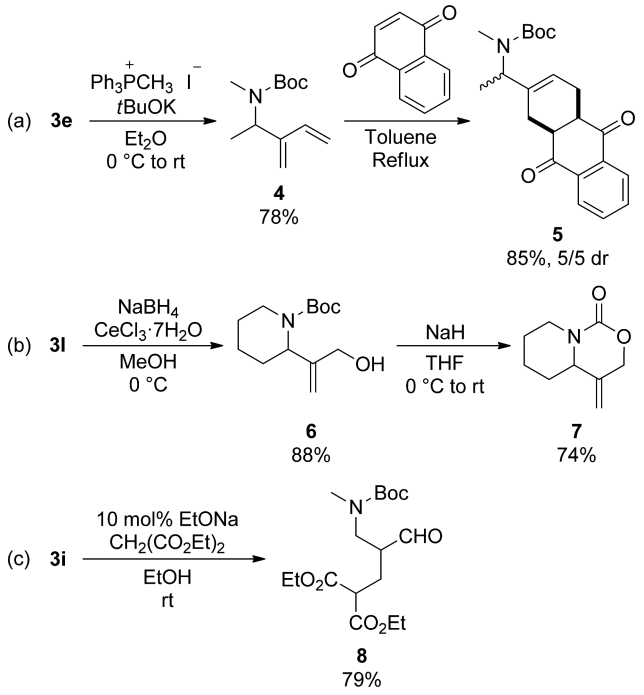
[a]Unless otherwise noted, the reactions were performed using approximately 0.2−0.8 mmol of **2**, and isolated yields of **3** are shown. [b]In parentheses are the yields obtained from over 1 mmol scale experiments (**2e**, 1.49 mmol; **2i**, 1.14 mmol).

by TLC analysis. The exact reason is currently unclear: the terminal dimethylallyl (prenyl) substituent might be protonated by Tf_2NH, with the resulting carbocation causing side reactions. Finally, we attempted the reactions of *N*-heterocyclic derivatives **2k** and **2l** and obtained the corresponding *β*-amino-*α*-methylene aldehydes **3k** (94%) and **3j** (93%). These compounds may act as valuable building blocks for the synthesis of alkaloid analogues.

With these results in hand, further synthetic transformations of products **3** were demonstrated (Scheme 4). Wittig olefination of **3e** with methylene ylide provided 2-substituted-1,3-butadiene **4** in a 78% yield, and Diels−Alder reaction with 1,4-naphthoquinone gave the corresponding adduct **5** in an 85% yield (Scheme 4a). Cyclic product **3l** could be transformed into 1,3-oxazine derivative **7** by 1,2-hydride reduction into **6** in an 88% yield, followed by base-promoted cyclization in a 74% yield (Scheme 4b). We attempted the same transformation using acyclic **3a**; however, the cyclization resulted in failure. Finally, the Michael addition of diethyl malonate into *α*-methylene aldehyde **3i** afforded *α*-substituted-*β*-aminoaldehyde derivative **8** in a 79% yield (Scheme 4c).

In conclusion, the Tf_2NH-catalyzed aza-Ferrier reaction of *N*,*O*-allenyl acetals **2** is reported. This protocol provides various types of *β*-amino-*α*-methylene aldehydes **3** as the products. Products **3** act as valuable building blocks, and further synthetic applications were successfully demonstrated, such as Wittig olefination followed by the Diels−Alder reaction, 1,2-reduction followed by cyclization, and Michael

Scheme 4. Further Synthetic Transformations of 3



addition. The substrates *N*,*O*-allenyl acetals **2** were easily prepared by base-induced isomerization of *N*,*O*-propargyl acetals **1** with Triton B. Our protocol would expand the product library of *β*-amino-*α*-methylene aldehydes.

EXPERIMENTAL SECTION

General. Infrared spectra (IR) were recorded on a JASCO FT/IR-4600 spectrometer. ^1H and ^{13}C NMR spectra were measured on a Varian (^1H, 400 MHz; ^{13}C, 100 MHz) or a Bruker (^1H, 400 MHz; ^{13}C, 100 MHz) spectrometer. As an internal standard in $CDCl_3$, Me_4Si (δ 0 ppm) for ^1H NMR and $CDCl_3$ (δ 77.00 ppm) for ^{13}C NMR were used. As an internal standard in acetone-d_6, the residual protons (δ 2.05 ppm) for ^1H NMR and acetone-d_6 (δ 29.92 ppm) for ^{13}C NMR were used. The splitting patterns are denoted as follows: s, singlet; d, doublet; t, triplet; q, quartet; m, multiplet; br, broad peak. High-resolution mass spectra (ESI) were measured on a Thermo Fisher Scientific LC/FT-MS spectrometer. Reactions involving air- or moisture-sensitive compounds were conducted in appropriate round-bottomed flasks with a magnetic stirring bar under an argon (Ar) atmosphere. Reactions under lower temperatures were carried out using a constant temperature bath with a magnetic stirrer (PSL-1800, EYELA, Japan) and an ultracooling reactor (UCR-150, Techno-Sigma Co., Ltd., Japan). Benzyltrimethylammonium hydroxide (Triton B) was purchased from Tokyo Chemical Industry Co., Ltd. (TCI) as a 40% MeOH solution and used after removal of MeOH under a vacuum. Trifluoromethanesulfonimide (Tf_2NH) was purchased from FUJIFILM Wako Chemical Corporation (Japan) and used without any purifications. Dichloromethane (CH_2Cl_2) was purchased from FUJIFILM Wako Chemical Corporation (Japan) and dried over 4 Å molecular sieves. For the thin-layer chromatography (TLC) analysis throughout this work, silica gel 70 TLC Plate-Wako purchased from FUJIFILM Wako Chemical Corporation (Japan) was used. The products were purified by preparative column chromatography on silica gel (60 N, spherical neutral) purchased from KANTO Chemical Co., Inc. (Japan).

Representative Procedure for Preparation of ***1a****.*[12] (Step 1) A mixture of *tert*-butyl carbamate (469 mg, 4.00 mmol), 3-phenylpropionaldehyde (0.58 mL, 4.4 mmol), sodium benzenesulfinate dihydrate (0.80 g, 6.0 mmol), and HCO_2H (1.0 mL, 27 mmol) in THF (1.6 mL) and H_2O (4.0 mL) was stirred for 18 h at room

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temperature. The resulting mixture was quenched with saturated aqueous $NaHCO_3$ at 0 °C and extracted with CH_2Cl_2. The combined extracts were washed with brine, dried over Na_2SO_4, and concentrated by evaporation to obtain *tert*-butyl (3-phenyl-1-(phenylsulfonyl)-propyl) carbamate (**9a**) (1.48 g, 99% yield) as a white solid. The crude product was used without purification in the next step. (Step 2) A solution of 2-propyn-1-ol (201 μL, 3.41 mmol) in THF (34 mL) was treated with NaH (60 wt % in oil, 0.16 g, 4.0 mmol) at 0 °C, and the mixture was stirred for 30 min at room temperature under an Ar atmosphere. The mixture was cooled at 0 °C, and a solid of **9a** (1.40 g, 3.73 mmol) was added. After stirring for 1 h at room temperature, the resulting mixture was quenched with saturated aqueous NH_4Cl at 0 °C. The mixture was extracted with EtOAc, and the combined extracts were washed with brine, dried over Na_2SO_4, and concentrated by evaporation. Purification of the residue by chromatography on silica gel (*n*-hexane/EtOAc = 6:1 to 5:1 as the eluent) gave *tert*-butyl (3-phenyl-1-(prop-2-yn-1-yloxy)pro-pyl)carbamate (**10a**) (0.92 g, 93% yield) as a colorless viscous oil. (Step 3) A solution of **10a** (0.92 g, 3.2 mmol) and MeI (0.30 mL, 4.8 mmol) in DMF (11 mL) was treated with NaH (60 wt % in oil, 0.19 g, 4.8 mmol) at 0 °C, and the mixture was stirred for 1 h at room temperature under an Ar atmosphere. The resulting mixture was cooled at 0 °C and quenched with saturated aqueous NH_4Cl at 0 °C. The mixture was extracted with EtOAc, and the combined extracts were washed with brine, dried over Na_2SO_4, and concentrated by evaporation. Purification of the residue by chromatography on silica gel (*n*-hexane/EtOAc = 6:1 to 4:1 as the eluent) gave *tert*-butyl methyl(3-phenyl-1-(prop-2-yn-1-yloxy)propyl)carbamate (**1a**) (840 mg, 87% yield) as a colorless viscous oil. ^1H NMR ($CDCl_3$, 400 MHz): 6:4 mixture of rotamers, δ 1.42 (s, 5.4H), 1.48 (s, 3.6H), 1.77−1.94 (m, 1H), 1.95−2.12 (m, 1H), 2.39 (dd, 0.6H, J = 2.0, 2.0 Hz), 2.41 (t, 0.4H, J = 2.0 Hz), 2.50−2.78 (m, 2H), 2.69 (s, 1.2H), 2.73 (s, 1.8H), 3.98 (dd, 0.6H, J = 15.8, 2.0 Hz), 4.08 (d, 0.8H, J = 2.0 Hz), 4.09 (dd, 0.6H, J = 15.8, 2.0 Hz), 5.40 (dd, 0.6H, J = 6.8, 6.8 Hz), 5.53 (dd, 0.4H, J = 6.8, 6.8 Hz), 7.14−7.23 (m, 3H), 7.23−7.32 (m, 2H). $^{13}C\{^1H\}$ NMR ($CDCl_3$, 100 MHz): 6:4 mixture of rotamers, δ 26.0, 27.0, 28.2, 28.3, 31.0, 31.4, 34.1, 34.5, 54.3, 55.2, 73.6, 73.9, 79.7, 79.8, 79.9, 80.3, 83.7, 84.3, 125.87, 125.93, 128.3, 140.9, 141.3, 155.2, 155.9. IR (ATR): 3290, 3250, 3027, 2975, 2930, 2862, 1691, 1496, 1478, 1454, 1425, 1402, 1392, 1365, 1336, 1257, 1167, 1138, 1100, 1058, 1041, 1024, 950, 910, 874, 836, 814, 770, 749, 728, 698 cm^{-1}. HRMS (ESI) m/z: $[M + Na]^+$ calcd for $C_{18}H_{25}NO_3Na$, 326.1727; found, 326.1725.

Representative Procedure for the Preparation of ***2a*** *by Isomerization of* ***1a*** *(Scheme 2).* A 40% benzyltrimethylammonium hydroxide (Triton B) MeOH solution (125 mg, 0.30 mmol) was charged in a flask and concentrated by evaporation. The residue was dried under reduced pressure (<1 mmHg) for 30 min. The flask was filled with Ar, and the residue was dissolved in DMSO (0.6 mL). A solution of **1a** (303 mg, 1.00 mmol) in DMSO (0.6 mL) was added to the solution, and the mixture was stirred for 16 h at room temperature. The resulting mixture was diluted with H_2O and extracted with a mixture of *n*-hexane/EtOAc = 3:1. The combined extracts were washed with brine, dried over Na_2SO_4, and concentrated by evaporation. The residue was purified by chromatography on silica gel (*n*-hexane/EtOAc = 15:1 as the eluent) to obtain *tert*-butyl methyl(3-phenyl-1-(propa-1,2-dien-1-yloxy)propyl)carbamate (**2a**) (254 mg, 84% yield) as a colorless oil. ^1H NMR ($CDCl_3$, 400 MHz): 5:5 mixture of rotamers, δ 1.40 (s, 4.5H), 1.47 (s, 4.5H), 1.90 (ddt, 1H, J = 14.0, 10.0, 6.8 Hz), 2.10 (br, 1H), 2.52−2.82 (m, 2H), 2.69 (s, 1.5H,), 2.74 (s, 1.5H), 5.32 (d, 1H, J = 6.0 Hz), 5.33 (d, 1H, J = 6.0 Hz), 5.39 (br, 0.5H), 5.61 (br, 0.5H), 6.44−6.58 (br m, 1H), 7.14−7.23 (m, 3H), 7.23−7.32 (m, 2H). $^{13}C\{^1H\}$ NMR ($CDCl_3$, 100 MHz): 5:5 mixture of rotamers, δ 26.0, 26.8, 28.3, 31.0, 31.2, 34.0, 34.2, 80.0, 80.2, 84.2, 84.5, 89.1, 117.8, 126.0, 128.3, 128.4, 140.8, 141.2, 154.7, 155.2, 201.2. IR (ATR): 3062, 3027, 3005, 2975, 2930, 2865, 1695, 1604, 1496, 1478, 1454, 1442, 1399, 1392, 1365, 1333, 1254, 1168, 1137, 1075, 1011, 876, 846, 768, 749, 698 cm^{-1}. HRMS (ESI) m/z: $[M + Na]^+$ calcd for $C_{18}H_{25}NO_3Na$, 326.1727; found, 326.1720.

*tert-Butyl Benzyl(3-phenyl-1-(prop-2-yn-1-yloxy)propyl)-carbamate (****1b****).* The compound was prepared by the same procedure with **1a** using benzyl bromide instead of MeI in Step 3:[12] 290 mg, 75% yield from **10a**, colorless oil, purified by chromatography on silica gel (*n*-hexane/EtOAc = 15:1 as the eluent). ^1H NMR ($CDCl_3$, 400 MHz): 5:5 mixture of rotamers, δ 1.39 (s, 4.5H), 1.45 (s, 4.5H), 1.73−1.88 (m, 1H), 1.88−2.03 (m, 1H), 2.30−2.80 (m, 3H), 3.81−4.19 (m, 2H), 4.22−4.36 (m, 1H), 4.36−4.65 (m, 1H), 5.50 (s, 0.5H), 5.65 (s, 0.5H), 7.01 (s, 2H), 7.09−7.44 (m, 8H). $^{13}C\{^1H\}$ NMR ($CDCl_3$, 100 MHz): 5:5 mixture of rotamers, δ 28.2, 31.2, 31.6, 35.3, 35.5, 44.0, 45.0, 54.4, 55.5, 73.9, 74.0, 79.7, 80.5, 84.7, 85.1, 125.8, 126.6, 127.0, 127.8, 128.2, 128.3, 139.6, 140.9, 141.1, 155.7, 156.0. IR (ATR): 3300, 3062, 3028, 3004, 2976, 2930, 2862, 1691, 1604, 1496, 1477, 1454, 1406, 1391, 1380, 1365, 1338, 1250, 1157, 1119, 1054, 1029, 1003, 955, 909, 890, 873, 819, 770, 734, 697 cm^{-1}. HRMS (ESI) m/z: $[M + Na]^+$ calcd for $C_{24}H_{29}NO_3Na$, 402.2040; found, 402.2032.

*tert-Butyl Benzyl(3-phenyl-1-(propa-1,2-dien-1-yloxy)propyl)-carbamate (****2b****).* The compound was prepared by the same procedure with **2a**: 208 mg, 75% yield, colorless oil, purified by chromatography on silica gel (*n*-hexane/EtOAc = 15:1 as the eluent). ^1H NMR ($CDCl_3$, 400 MHz): 6:4 mixture of rotamers, δ 1.36 (br, 9H), 1.86 (br, 1H), 2.00 (br, 1H), 2.35−2.80 (br, 2H), 4.25 (s, 0.4H), 4.29 (s, 0.6H), 4.46 (br, 0.6H), 4.61 (br, 0.4H), 5.35 (d, 2H, J = 6.0 Hz), 5.48 (br, 0.4H), 5.73 (br, 0.6H), 6.36−6.67 (br, 1H), 7.02 (br, 2H), 7.15 (t, 1H, J = 7.2 Hz), 7.18−7.25 (m, 3H), 7.29 (br, 4H). $^{13}C\{^1H\}$ NMR ($CDCl_3$, 100 MHz): 6:4 mixture of rotamers, δ 28.2, 31.4, 35.4, 44.7, 80.5, 85.0, 89.4, 118.3, 125.9, 126.6, 127.0, 127.6, 128.2, 128.3, 128.4, 139.8, 141.0, 155.2, 201.1. IR (ATR): 3087, 3062, 3028, 3005, 2975, 2929, 2865, 1694, 1604, 1496, 1477, 1453, 1405, 1391, 1380, 1365, 1335, 1297, 1249, 1150, 1120, 1085, 1074, 1022, 957, 924, 881, 854, 819, 768, 737, 696 cm^{-1}. HRMS (ESI) m/z: $[M + Na]^+$ calcd for $C_{24}H_{29}NO_3Na$, 402.2040; found, 402.2030.

*tert-Butyl Allyl(3-phenyl-1-(prop-2-yn-1-yloxy)propyl)carbamate (****1c****).* The compound was prepared by the same procedure with **1a** using allyl bromide instead of MeI in Step 3:[12] 248 mg, 62% yield from **10a**, colorless oil, purified by chromatography on silica gel (*n*-hexane/EtOAc = 15:1 as the eluent). ^1H NMR ($CDCl_3$, 400 MHz): 5:5 mixture of rotamers, δ 1.42 (s, 4.5H), 1.47 (s, 4.5H), 1.79−1.94 (br m, 1H), 1.94−2.09 (br m, 1H), 2.41 (s, 1H), 2.52−2.82 (m, 2H), 3.61−3.92 (m, 2H), 3.92−4.15 (m, 2H), 5.02−5.24 (m, 2H), 5.41 (t, 0.5H, J = 5.8 Hz), 5.55 (t, 0.5H, J = 6.2 Hz), 5.73−6.01 (m, 1H), 7.12−7.22 (m, 3H), 7.22−7.32 (m, 2H). $^{13}C\{^1H\}$ NMR ($CDCl_3$, 100 MHz): 5:5 mixture of rotamers, δ 28.2, 31.3, 31.6, 35.0, 35.3, 43.2, 43.8, 54.4, 55.3, 73.7, 74.0, 79.7, 80.1, 80.4, 84.3, 84.7, 116.2, 125.8, 128.28, 128.33, 135.2, 135.5, 141.0, 141.3, 154.8, 155.7. IR (ATR): 3299, 3252, 3084, 3064, 3027, 3005, 2977, 2930, 2863, 1692, 1496, 1477, 1454, 1403, 1365, 1330, 1251, 1164, 1138, 1053, 996, 921, 913, 874, 771, 750, 738, 722, 698 cm^{-1}. HRMS (ESI) m/z: $[M + Na]^+$ calcd for $C_{20}H_{27}NO_3Na$, 352.1883; found, 352.1881.

*tert-Butyl Allyl(3-phenyl-1-(propa-1,2-dien-1-yloxy)propyl)-carbamate (****2c****).* The compound was prepared by the same procedure with **2a**: 173 mg, 71% yield, colorless oil, purified by chromatography on silica gel (*n*-hexane/EtOAc = 15:1 as the eluent), colorless oil. ^1H NMR ($CDCl_3$, 400 MHz): 6:4 mixture of rotamers, δ 1.45 (s, 9H), 1.91 (dddd, 1H, J = 13.6, 10.0, 6.4, 6.4 Hz), 2.08 (br, 1H), 2.56−2.79 (br, 1H), 2.61 (ddd, 1H, J = 13.9, 9.7, 6.4 Hz), 3.60−3.95 (br, 2H), 5.02−5.24 (br, 1H), 5.08 (d, 1H, J = 10.0 Hz), 5.34 (d, 2H, J = 5.6 Hz), 5.41 (br, 0.4H), 5.63 (br, 0.6H), 5.85 (br, 1H), 6.52 (br, 1H), 7.14−7.22 (m, 3H), 7.23−7.31 (m, 2H). $^{13}C\{^1H\}$ NMR ($CDCl_3$, 100 MHz): 6:4 mixture of rotamers, δ 28.3, 29.7, 31.3, 31.4, 35.2, 43.2, 43.6, 80.2, 80.3, 84.6, 85.0, 89.2, 115.8, 118.1, 125.9, 128.4, 135.5, 135.7, 141.0, 141.2, 154.2, 154.9, 201.1. IR (ATR): 3084, 3063, 3027, 3005, 2976, 2929, 2865, 1696, 1496, 1477, 1453, 1444, 1403, 1391, 1364, 1328, 1250, 1161, 1137, 1081, 1067, 1022, 911, 878, 846, 769, 750, 720, 698 cm^{-1}. HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{20}H_{28}NO_3$, 330.2064; found, 330.2058.

*tert-Butyl Methyl(1-(prop-2-yn-1-yloxy)pentyl)carbamate (****1d****).* The compound was prepared by the same procedure with **1a** using *n*-valeraldehyde instead of 3-phenylpropionaldehyde in Step 1:[12] 175

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mg, 54% overall yield, colorless oil, purified by chromatography on silica gel (*n*-hexane/EtOAc = 10:1 as the eluent). ^{1}H NMR ($CDCl_3$, 400 MHz): 6:4 mixture of rotamers, δ 0.84−0.97 (m, 3H), 1.15−1.75 (m, 6H), 1.48 (s, 9H), 2.41 (dd, 1H, J = 2.2, 2.2 Hz), 2.68 (s, 1.2H) 2.69 (s, 1.8H), 3.98 (dd, 0.6H, J = 15.6, 2.2 Hz), 4.04−4.15 (m, 1.4H), 5.42 (t, 0.6H, J = 6.6 Hz), 5.47 (t, 0.4H, J = 6.8 Hz). ^{13}C{^{1}H} NMR ($CDCl_3$, 100 MHz): 6:4 mixture of rotamers, δ 13.9, 22.1, 22.3, 25.8, 26.78, 26.82, 26.9, 28.3, 32.1, 32.2, 54.1, 55.0, 73.5, 73.7, 79.7, 79.8, 79.9, 80.1, 84.2, 84.6, 155.4, 155.9. IR (ATR): 3311, 3252, 2958, 2933, 2872, 2862, 1692, 1478, 1455, 1437, 1428, 1402, 1392, 1365, 1337, 1258, 1213, 1152, 1130, 1068, 1042, 1014, 943, 915, 874, 772, 733 cm^{-1}. HRMS (ESI) m/z: [M + Na]$^+$ calcd for $C_{14}H_{25}NO_3Na$, 278.1727; found, 278.1724.

*tert-Butyl Methyl(1-(propa-1,2-dien-1-yloxy)pentyl)carbamate (**2d**).* The compound was prepared by the same procedure with **2a**: 404 mg, 58% yield, colorless oil, purified by chromatography on silica gel (*n*-hexane/EtOAc = 15:1 as the eluent), colorless oil. ^{1}H NMR ($CDCl_3$, 400 MHz): 5:5 mixture of rotamers, δ 0.90 (t, 3H, J = 6.2 Hz), 1.16−1.42 (m, 4H), 1.47 (s, 9H), 1.53−1.82 (m, 2H), 2.69 (s, 1.5H), 2.70 (s, 1.5H), 5.26−5.46 (m, 2H), 5.40 (t, 0.5H, J = 6.6 Hz), 5.55 (t, 0.5H, J = 6.4 Hz), 6.46−6.56 (br m, 1H). ^{13}C{^{1}H} NMR ($CDCl_3$, 100 MHz): 5:5 mixture of rotamers, δ 13.9, 22.2, 22.3, 25.8, 26.7, 26.8, 26.9, 28.4, 31.9, 32.1, 79.8, 80.1, 84.5, 85.1, 88.9, 89.1, 117.8, 117.9, 154.8, 155.2, 201.1. IR (ATR): 3006, 2959, 2932, 2872, 2863, 1697, 1478, 1442, 1400, 1392, 1365, 1335, 1249, 1154, 1122, 1072, 1016, 939, 876, 847, 768, 735 cm^{-1}. HRMS (ESI) m/z: [M + Na]$^+$ calcd for $C_{14}H_{25}NO_3Na$, 278.1727; found, 278.1722.

*tert-Butyl Methyl(1-(prop-2-yn-1-yloxy)ethyl)carbamate (**1e**).* The compound was prepared by the same procedure with **1a** using acetaldehyde instead of 3-phenylpropionaldehyde in Step 1:[12] 711 mg, 62% overall yield, colorless crystals. Mp: 42−44 °C, purified by chromatography on silica gel (*n*-hexane/EtOAc = 10:1 to 5:1 as the eluent). ^{1}H NMR ($CDCl_3$, 400 MHz): 5:5 mixture of rotamers, δ 1.31 (d, 3H, J = 5.8 Hz), 1.49 (s, 9H), 2.42 (dd, 1H, J = 2.4, 2.4 Hz), 2.72 (s, 3H), 3.98 (d, 0.5H, J = 16.0 Hz), 4.02−4.14 (m, 1.5H), 5.60 (q, 0.5H, J = 5.8 Hz), 5.67 (q, 0.5H, J = 5.8 Hz). ^{13}C{^{1}H} NMR ($CDCl_3$, 100 MHz): 5:5 mixture of rotamers, δ 18.9, 19.1, 25.7, 26.6, 28.3, 54.1, 54.9, 73.5, 73.8, 79.68, 79.76, 79.82, 80.2, 80.7, 80.9, 154.9, 155.6. IR (ATR): 3240, 2980, 2969, 2932, 2913, 2865, 1677, 1477, 1432, 1405, 1393, 1379, 1365, 1352, 1261, 1154, 1125, 1080, 1061, 1040, 1028, 984, 925, 872, 849, 817, 777, 763, 740, 688 cm^{-1}. HRMS (ESI) m/z: [M + Na]$^+$ calcd for $C_{11}H_{19}NO_3Na$, 236.1257; found, 236.1256.

*tert-Butyl Methyl(1-(propa-1,2-dien-1-yloxy)ethyl)carbamate (**2e**).* The compound was prepared by the same procedure with **2a**: 390 mg, 76% yield, colorless oil, purified by chromatography on silica gel (*n*-hexane/EtOAc = 15:1 to 10:1 as the eluent). ^{1}H NMR (acetone-d_6, 400 MHz): δ 1.30 (br s, 3H), 1.45 (s, 9H), 2.69 (s, 3H), 5.31 (br, 2H), 5.48−5.78 (br m, 1H), 6.57 (br, 1H); ^{13}C{^{1}H} NMR (acetone-d_6, 100 MHz): 5:5 mixture of rotamers, δ 18.7, 18.9, 26.2, 26.9, 28.6, 80.2, 81.8, 82.7, 89.0, 118.6, 154.7, 155.5, 202.5. IR (ATR): 2978, 2934, 1696, 1479, 1441, 1401, 1392, 1366, 1331, 1256, 1163, 1151, 1114, 1074, 1011, 869, 847, 770, 714, 689 cm^{-1}. HRMS (ESI) m/z: [M + H]$^+$ calcd for $C_{11}H_{20}NO_3$, 214.1438; found, 214.1437.

*tert-Butyl Methyl(2-phenyl-1-(prop-2-yn-1-yloxy)ethyl)carbamate (**1f**).* The compound was prepared by the same procedure with **1a** using phenylacetaldehyde instead of 3-phenylpropionaldehyde in Step 1:[12] 141 mg, 22% overall yield, colorless oil, purified by chromatography on silica gel (*n*-hexane/EtOAc = 10:1 as the eluent). ^{1}H NMR ($CDCl_3$, 400 MHz): 6:4 mixture of rotamers, δ 1.28 (s, 5.4H), 1.38 (s, 3.6H), 2.36−2.42 (m, 1H), 2.73 (s, 1.2H), 2.78 (s, 1.8H), 2.79−2.91 (m, 1H), 2.94−3.10 (m, 1H), 4.00 (dd, 0.6H, J = 15.9, 2.4 Hz), 4.07 (d, 0.8H, J = 2.4 Hz), 4.11 (dd, 0.6H, J = 15.9, 2.4 Hz), 5.67 (dd, 0.6H, J = 8.4, 5.2 Hz), 5.78 (dd, 0.4H, J = 6.8, 6.8 Hz), 7.14−7.31 (m, 5H). ^{13}C{^{1}H} NMR ($CDCl_3$, 100 MHz): 6:4 mixture of rotamers, δ 26.1, 27.1, 28.0, 28.2, 39.0, 39.1, 54.4, 55.3, 73.7, 74.1, 79.3, 79.6, 79.8, 80.0, 84.9, 85.2, 126.4, 126.5, 128.2, 128.3, 129.1, 136.3, 136.4, 154.9, 155.6. IR (ATR): 3302, 3251, 3030, 3005, 2976, 2931, 2863, 1691, 1496, 1478, 1455, 1427, 1392, 1365, 1351, 1296, 1257, 1168, 1139, 1093, 1055, 1019, 959, 918, 876, 823, 771, 735, 698 cm^{-1}. HRMS (ESI) m/z: [M + Na]$^+$ calcd for $C_{17}H_{23}NO_3Na$, 312.1570; found, 312.1565.

*tert-Butyl Methyl(2-phenyl-1-(propa-1,2-dien-1-yloxy)ethyl)carbamate (**2f**).* The compound was prepared by the same procedure with **2a**: 84 mg, 31% yield, colorless oil, purified by chromatography on silica gel (*n*-hexane/EtOAc = 15:1 as the eluent). ^{1}H NMR ($CDCl_3$, 400 MHz): 6:4 mixture of rotamers, δ 1.21 (s, 5.4H), 1.38 (s, 3.6H), 2.73 (s, 1.2H), 2.79 (s, 1.8H), 2.83−2.95 (m, 0.4H), 2.90 (dd, 0.6H, J = 13.6, 8.3 Hz), 3.03 (dd, 0.6H, J = 13.6, 5.1 Hz), 3.10 (dd, 0.4H, J = 13.8, 6.9 Hz), 5.26−5.40 (m, 2H), 5.58 (dd, 0.6H, J = 8.3, 5.1 Hz), 5.80 (dd, 0.4H, J = 6.9, 6.4 Hz), 6.52 (t, 1H, J = 6.0 Hz), 7.13−7.32 (m, 5H). ^{13}C{^{1}H} NMR ($CDCl_3$, 100 MHz): 6:4 mixture of rotamers, δ 26.0, 27.0, 28.0, 28.3, 38.8, 38.9, 79.9, 84.6, 85.4, 89.3, 89.6, 117.6, 117.9, 126.5, 126.6, 128.2, 128.4, 129.3, 129.4, 136.1, 136.2, 154.3, 154.7, 200.8. IR (ATR): 3030, 3005, 2976, 2930, 1695, 1496, 1478, 1442, 1392, 1365, 1335, 1295, 1256, 1168, 1137, 1092, 1065, 1028, 1011, 956, 881, 849, 766, 747, 729, 698 cm^{-1}. HRMS (ESI) m/z: [M + H]$^+$ calcd for $C_{17}H_{24}NO_3$, 290.1751; found, 290.1744.

*tert-Butyl Methyl(phenyl(prop-2-yn-1-yloxy)methyl)carbamate (**1g**).* The compound was prepared by the same procedure with **1a** using benzaldehyde instead of 3-phenylpropionaldehyde in Step 1:[12] 136 mg, 58% overall yield, colorless oil, purified by chromatography on silica gel (*n*-hexane/EtOAc = 10:1 as the eluent). ^{1}H NMR ($CDCl_3$, 400 MHz): 6:4 mixture of rotamers, δ 1.53 (s, 3.6H), 1.55 (s, 5.4H), 2.44 (s, 1H), 2.54 (s, 3H), 4.21 (d, 0.6H, J = 16.0 Hz), 4.25−4.39 (m, 1.4H), 6.60 (s, 0.6H), 6.66 (s, 0.4H), 7.25−7.43 (m, 5H). ^{13}C{^{1}H} NMR ($CDCl_3$, 100 MHz): 6:4 mixture of rotamers, δ 27.2, 28.0, 28.3, 54.4, 55.4, 73.9, 74.3, 79.4, 79.6, 80.2, 80.7, 84.2, 84.3, 126.1, 126.2, 127.9, 128.0, 128.2, 128.3, 137.9, 138.0, 155.5, 156.5. IR (ATR): 3295, 3250, 3063, 3033, 3006, 2977, 2931, 2864, 1689, 1477, 1450, 1439, 1428, 1391, 1366, 1326, 1257, 1137, 1081, 1063, 1042, 1025, 958, 916, 884, 866, 849, 768, 739, 718, 695 cm^{-1}. HRMS (ESI) m/z: [M + Na]$^+$ calcd for $C_{16}H_{21}NO_3Na$, 298.1414; found, 298.1408.

*tert-Butyl Methyl(phenyl(propa-1,2-dien-1-yloxy)methyl)carbamate (**2g**).* The compound was prepared by the same procedure with **2a**: 92 mg, 67% yield, colorless oil, purified by chromatography on silica gel (*n*-hexane/EtOAc = 15:1 as the eluent). ^{1}H NMR ($CDCl_3$, 400 MHz): 5:5 mixture of rotamers, δ 1.52 (s, 9H), 2.58 (br s, 3H), 5.38 (br, 2H), 6.54 (br, 0.5H), 6.64−6.77 (br, 1.5H), 7.26−7.45 (m, 5H). ^{13}C{^{1}H} NMR ($CDCl_3$, 100 MHz): 5:5 mixture of rotamers, δ 27.0, 27.7, 28.4, 80.2, 80.6, 84.0, 84.8, 89.6, 89.8, 117.9, 126.0, 126.2, 128.0, 128.1, 128.3, 137.5, 154.8, 155.6, 201.1. IR (ATR): 3033, 3006, 2976, 2931, 1693, 1477, 1439, 1390, 1366, 1323, 1255, 1172, 1143, 1079, 1018, 955, 923, 886, 850, 767, 743, 716, 696 cm^{-1}. HRMS (ESI) m/z: [M + Na]$^+$ calcd for $C_{16}H_{21}NO_3Na$, 298.1414; found, 298.1409.

*tert-Butyl Methyl(2-methyl-1-(prop-2-yn-1-yloxy)propyl)carbamate (**1h**).* The compound was prepared by the same procedure with **1a** using isobutyraldehyde instead of 3-phenylpropionaldehyde in Step 1:[12] 198 mg, 78% overall yield, colorless oil, purified by chromatography on silica gel (*n*-hexane/EtOAc = 15:1 as the eluent). ^{1}H NMR ($CDCl_3$, 400 MHz): 6:4 mixture of rotamers, δ 0.80 (d, 1.2H, J = 6.8 Hz), 0.81 (d, 1.8H, J = 6.8 Hz), 1.02 (d, 1.8H, J = 6.8 Hz), 1.04 (d, 1.2H, J = 6.8 Hz), 1.47 (s, 3.6H), 1.48 (s, 5.4H), 1.77−1.94 (m, 1H), 2.37−2.42 (m, 1H), 2.67 (s, 1.2H), 2.69 (s, 1.8H), 3.97 (dd, 0.6H, J = 15.6, 2.4 Hz), 4.06 (d, 0.8H, J = 2.0 Hz), 4.10 (dd, 0.6H, J = 16.0, 2.0 Hz), 4.96 (d, 0.6H, J = 9.6 Hz), 5.05 (d, 0.4H, J = 10.0 Hz). ^{13}C{^{1}H} NMR ($CDCl_3$, 100 MHz): 6:4 mixture of rotamers, δ 17.5, 17.9, 19.09, 19.11, 26.2, 27.1, 28.3, 30.49, 30.54, 54.5, 55.3, 73.5, 73.7, 79.7, 79.9, 80.1, 89.8, 89.9, 155.7, 156.3. IR (ATR): 3311, 3251, 2965, 2931, 2876, 1692, 1473, 1439, 1427, 1400, 1389, 1366, 1336, 1311, 1259, 1160, 1136, 1065, 1042, 1024, 974, 936, 908, 876, 841, 770, 736 cm^{-1}. HRMS (ESI) m/z: [M + Na]$^+$ calcd for $C_{13}H_{23}NO_3Na$, 264.1570; found, 264.1568.

*tert-Butyl Methyl(2-methyl-1-(propa-1,2-dien-1-yloxy)propyl)carbamate (**2h**).* The compound was prepared by the same procedure with **2a**: 91 mg, 65% yield, colorless oil, purified by

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chromatography on silica gel (*n*-hexane/EtOAc = 15:1 as the eluent). ^{1}H NMR ($CDCl_3$, 400 MHz): 5:5 mixture of rotamers, δ 0.83 (d, 3H, J = 6.8 Hz), 1.02 (d, 1.5H, J = 5.2 Hz), 1.03 (d, 1.5H, J = 6.0 Hz), 1.47 (s, 9H), 1.87−2.05 (m, 1H), 2.67 (s, 1.5H), 2.70 (s, 1.5H), 4.97 (d, 0.5H, J = 9.2 Hz), 5.14 (d, 0.5H, J = 9.2 Hz), 5.26−5.41 (m, 2H), 6.47−6.56 (m, 1H). ^{13}C{^{1}H} NMR ($CDCl_3$, 100 MHz): 5:5 mixture of rotamers, δ 17.4, 17.7, 18.77, 18.81, 26.1, 26.9, 28.3, 28.4, 30.6, 31.0, 79.6, 79.9, 88.9, 89.1, 89.3, 90.0, 118.06, 118.08, 154.9, 155.4, 200.99, 201.01. IR (ATR): 2967, 2931, 2876, 1697, 1472, 1441, 1399, 1389, 1365, 1331, 1310, 1292, 1257, 1160, 1127, 1019, 961, 946, 933, 920, 877, 846, 768, 738, 718 cm^{-1}. HRMS (ESI) m/z: [M + H]$^+$ calcd for $C_{13}H_{24}NO_3$, 242.1751; found, 242.1742.

*tert-Butyl Methyl((prop-2-yn-1-yloxy)methyl)carbamate (**1i**).* (Step 1)[13] A mixture of *tert*-butyl carbamate (0.75 g, 6.4 mmol), 37 wt % HCHO H_2O solution (1.0 mL, ca. 13 mmol), sodium benzenesulfinate dihydrate (2.56 g, 12.8 mmol), and HCO_2H (0.75 mL, 20 mmol) in MeOH (8 mL) and H_2O (16 mL) was stirred for 5 days at room temperature. Colorless crystals that formed in the resulting mixture were isolated by filtration and washed with Et_2O. The crystals were dissolved in CH_2Cl_2, and the solution was dried over Na_2SO_4. The solution was concentrated by evaporation to obtain *tert*-butyl ((phenylsulfonyl)methyl)carbamate (**9i**) (826 mg, 48% yield) as a white solid. The crude product was used without purification in the next step. The following procedure was the same as that with **1a**:[12] 368 mg, 67% overall yield from **9i**, colorless oil, purified by chromatography on silica gel (*n*-hexane/EtOAc = 7:1 to 5:1 as the eluent). ^{1}H NMR ($CDCl_3$, 400 MHz): 6:4 mixture of rotamers, δ 1.49 (s, 9H), 2.43 (t, 1H, J = 2.4 Hz), 2.91 (s, 3H), 4.12 (br, 1.2H), 4.15 (br, 0.8H), 4.82 (s, 2H). ^{13}C{^{1}H} NMR ($CDCl_3$, 100 MHz): 6:4 mixture of rotamers, δ 28.2, 32.9, 33.5, 54.5, 55.4, 73.8, 73.9, 77.8, 78.1, 79.65, 79.74, 80.1, 80.4, 155.0, 155.9. IR (ATR): 3292, 3254, 2977, 2933, 1695, 1480, 1434, 1389, 1366, 1301, 1254, 1207, 1154, 1074, 1043, 987, 932, 888, 872, 773, 692 cm^{-1}. HRMS (ESI) m/z: [M + Na]$^+$ calcd for $C_{10}H_{17}NO_3Na$, 222.1101; found, 222.1099.

*tert-Butyl Methyl((propa-1,2-dien-1-yloxy)methyl)carbamate (**2i**).* The compound was prepared by the same procedure with **2a**: 228 mg, 62% yield, colorless oil, purified by chromatography on silica gel (*n*-hexane/EtOAc = 15:1 to 10:1 as the eluent). ^{1}H NMR ($CDCl_3$, 400 MHz): 5:5 mixture of rotamers, δ 1.48 (s, 9H), 2.93 (s, 3H), 4.81 (s, 1H), 4.83 (s, 1H), 5.40 (d, 2H, J = 5.6 Hz), 6.67 (br, 1H). ^{13}C{^{1}H} NMR ($CDCl_3$, 100 MHz): 5:5 mixture of rotamers, δ 28.3, 33.5, 33.8, 77.9, 78.3, 80.5, 89.9, 118.8, 154.8, 155.5, 200.9. IR (ATR): 3005, 2977, 2932, 1701, 1480, 1443, 1386, 1366, 1348, 1291, 1250, 1146, 1067, 1016, 948, 923, 876, 852, 773, 733, 698 cm^{-1}. HRMS (ESI) m/z: [M + Na]$^+$ calcd for $C_{10}H_{17}NO_3Na$, 222.1101; found, 222.1100.

*tert-Butyl ((3R)-3,7-Dimethyl-1-(prop-2-yn-1-yloxy)oct-6-en-1-yl)(methyl)carbamate (**1j**).* The compound was prepared by the same procedure with **1a** using (+)-citronellal instead of 3-phenylpropionaldehyde in Step 1:[12] 877 mg, 72% overall yield, colorless oil, purified by chromatography on silica gel (*n*-hexane/EtOAc = 10:1 to 8:1 as the eluent). ^{1}H NMR ($CDCl_3$, 400 MHz): mixture of rotamers and diastereomers, δ 0.93 (d, 3H, J = 6.8 Hz), 1.05−1.80 (m, 14H), 1.60 (s, 3H), 1.68 (s, 3H), 1.85−2.10 (m, 2H), 2.41 (dd, 1H, J = 2.3, 2.3 Hz), 2.65−2.71 (m, 3H), 3.95 (dd, 0.2H, J = 5.8, 2.3 Hz), 3.99 (dd, 0.4H, J = 5.8, 2.3 Hz), 4.03−4.14 (m, 1.4H), 5.04−5.12 (m, 1H), 5.50−5.63 (m, 1H). ^{13}C{^{1}H} NMR ($CDCl_3$, 100 MHz): mixture of rotamers and diastereomers, δ 17.56, 17.59, 19.2, 19.4, 19.5, 19.6, 25.22, 25.24, 25.28, 25.31, 25.6, 25.7, 26.0, 26.7, 27.0, 28.2, 28.3, 28.66, 28.74, 28.8, 28.9, 36.6, 36.8, 37.1, 37.2, 39.2, 39.3, 39.8, 39.9, 54.0, 54.1, 54.9, 73.5, 73.7, 79.69, 79.71, 79.9, 80.1, 80.2, 82.7, 83.0, 83.2, 124.4, 124.51, 124.54, 124.6, 130.96, 131.05, 131.08, 131.2, 155.2, 155.3, 155.7, 155.8. IR (ATR): 3311, 3253, 2967, 2927, 2857, 1695, 1477, 1454, 1438, 1426, 1403, 1391, 1365, 1337, 1257, 1150, 1133, 1072, 1049, 1017, 945, 909, 875, 828, 771, 732, 686 cm^{-1}. HRMS (ESI) m/z: [M + Na]$^+$ calcd for $C_{19}H_{33}NO_3Na$, 346.2353; found, 346.2349.

*tert-Butyl ((3R)-3,7-Dimethyl-1-(propa-1,2-dien-1-yloxy)oct-6-en-1-yl)(methyl)carbamate (**2j**).* The compound was prepared by the same procedure with **2a**: 265 mg, 82% yield, colorless oil, purified by chromatography on silica gel (*n*-hexane/EtOAc = 20:1 as the eluent). ^{1}H NMR (acetone-d_6, 400 MHz): mixture of diastereomers and rotamers, δ 0.90−1.00 (m, 3H), 1.07−1.56 (m, 4H), 1.46 (s, 9H), 1.60 (s, 3H), 1.63−1.83 (m, 1H), 1.66 (d, 3H, J = 1.4 Hz), 1.89−2.12 (m, 2H), 2.67 (br s, 3H), 5.10 (tq, 1H, J = 7.2, 1.4 Hz), 5.25−5.43 (m, 1H), 5.30 (d, 1H, J = 5.3 Hz), 5.55 (dd, 0.5H, J = 12.4, 5.6 Hz), 5.67 (dd, 0.5H, J = 11.8, 5.6 Hz), 6.57 (dd, 1H, J = 5.3, 5.3 Hz). ^{13}C{^{1}H} NMR (acetone-d_6, 100 MHz): mixture of diastereomers and rotamers, δ 17.8, 19.9, 20.2, 26.0, 26.1, 26.3, 26.4, 27.1, 27.2, 28.6, 37.47, 37.53, 38.0, 39.7, 40.0, 40.2, 40.5, 80.2, 80.37, 80.43, 83.9, 84.1, 84.7, 84.8, 89.0, 89.1, 89.4, 118.6, 118.7, 125.5, 131.7, 131.8, 154.9, 155.0, 155.6, 155.8, 202.5. IR (ATR): 2968, 2927, 2855, 1698, 1478, 1442, 1401, 1391, 1365, 1333, 1255, 1170, 1149, 1129, 1068, 1018, 941, 908, 879, 849, 768, 742, 715, 697 cm^{-1}. HRMS (ESI) m/z: [M + H]$^+$ calcd for $C_{19}H_{34}NO_3$, 324.2533; found, 324.2529.

*Representative Procedure for Preparation of **1k**.*[12] (Step 1) A solution of 2-pyrrolidinone (0.38 mL, 5.0 mmol) in THF (12.5 mL) was treated with a 1.6 M *n*BuLi hexane solution (3.1 mL, 5.0 mmol) at 0 °C under an Ar atmosphere, and the mixture was stirred for 30 min at the same temperature. The mixture was treated with $(Boc)_2O$ (1.26 mL, 5.5 mmol) at 0 °C and stirred for 2 h at room temperature. The resulting mixture was quenched with saturated aqueous NH_4Cl and extracted with EtOAc. The combined extracts were washed with brine, dried over Na_2SO_4, and concentrated by evaporation to obtain *tert*-butyl 2-oxopyrrolidine-1-carboxylate (**11k**) as a colorless oil. The crude product was used without purification in next step. (Step 2) A 1.0 M sodium bis(2-methoxyethoxy)aluminum hydride toluene solution (6.0 mL, 6.0 mmol) was added to a solution of **11k** in CH_2Cl_2 (25 mL) at −78 °C under an Ar atmosphere, and the mixture was stirred for 2 h at the same temperature. The resulting mixture was quenched with MeOH (2.5 mL) at −78 °C and treated with a saturated aqueous solution of potassium sodium tartrate (25 mL). The mixture was stirred at room temperature for a few hours until a clear two-phase solution was obtained. The resulting mixture was diluted with H_2O and extracted with CH_2Cl_2. The combined extracts were dried over Na_2SO_4 and concentrated by evaporation. The residue was purified by chromatography on silica gel (*n*-hexane/EtOAc = 3:1 to 1.5:1 as the eluent) to obtain *tert*-butyl 2-hydroxypyrrolidine-1-carboxylate (**12k**) (633 mg, 68% yield from Step 1) as a colorless oil. (Step 3) $Sc(OTf)_3$ (33 mg, 0.067 mmol) was added to a mixture of **12k** (633 mg, 3.38 mmol), 2-propyn-1-ol (0.24 mL, 4.1 mmol), and 3 Å molecular sieves (1.4 g) in CH_2Cl_2 (17 mL), and the mixture was stirred for 18 h at room temperature under an Ar atmosphere. The resulting mixture was treated with saturated aqueous $NaHCO_3$ and extracted with CH_2Cl_2. The combined extracts were washed with brine, dried over Na_2SO_4, and concentrated by evaporation. Purification of the residue by chromatography on silica gel (*n*-hexane/EtOAc = 6:1 to 5:1 as the eluent) gave *tert*-butyl 2-(prop-2-yn-1-yloxy)pyrrolidine-1-carboxylate (**1k**) (530 mg, 70% yield) as colorless crystals. Mp: 47−49 °C. ^{1}H NMR ($CDCl_3$, 400 MHz): 5:5 mixture of rotamers, δ 1.48 (s, 4.5H), 1.50 (s, 4.5H), 1.73−1.94 (m, 2H), 1.94−2.17 (m, 2H), 2.40 (t, 1H, J = 2.4 Hz), 3.22−3.47 (m, 2H), 4.21 (s, 1H), 4.30 (s, 1H), 5.33 (s, 0.5H), 5.39 (s, 0.5H). ^{13}C{^{1}H} NMR ($CDCl_3$, 100 MHz): 5:5 mixture of rotamers, δ 21.6, 22.6, 28.3, 32.5, 33.1, 45.4, 46.0, 55.1, 56.0, 73.3, 73.6, 79.8, 80.3, 80.4, 80.6, 86.8, 87.1, 154.1, 155.2. IR (ATR): 3234, 2992, 2974, 2958, 2933, 2896, 1681, 1478, 1456, 1443, 1382, 1366, 1343, 1328, 1292, 1269, 1255, 1165, 1123, 1101, 1049, 1027, 975, 934, 910, 877, 849, 832, 787, 774, 750, 693 cm^{-1}. HRMS (ESI) m/z: [M + Na]$^+$ calcd for $C_{12}H_{19}NO_3Na$, 248.1257; found, 248.1255.

*tert-Butyl 2-(Propa-1,2-dien-1-yloxy)pyrrolidine-1-carbo-xylate (**2k**).* The compound was prepared by the same procedure with **2a** (141 mg, 76% yield, colorless oil) and purified by chromatography on silica gel (*n*-hexane/EtOAc = 10:1 to 6:1 as the eluent). ^{1}H NMR (acetone-d_6, 400 MHz): 5:5 mixture of rotamers, δ 1.45 (s, 9H), 1.70−2.10 (m, 4H), 3.16−3.35 (m, 1H), 3.35−3.50 (m, 1H), 5.25−5.47 (m, 3H), 6.74 (t, 0.5H, J = 5.4 Hz), 6.82 (t, 0.5H, J = 5.4 Hz). ^{13}C{^{1}H} NMR ($CDCl_3$, 100 MHz): 5:5 mixture of rotamers, δ 22.2,

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23.1, 28.6, 32.3, 33.1, 46.4, 47.0, 80.1, 88.5, 88.7, 89.0, 119.7, 154.0, 154.8, 202.9. IR (ATR): 2976, 2933, 2888, 1698, 1478, 1455, 1443, 1382, 1365, 1341, 1322, 1286, 1255, 1155, 1119, 1094, 1030, 1015, 975, 946, 931, 909, 882, 851, 826, 772 cm^{-1}. HRMS (ESI) *m/z*: $[M + H]^+$ calcd for $C_{12}H_{20}NO_3$, 226.1438; found, 226.1433.

*tert-Butyl 2-(Prop-2-yn-1-yloxy)piperidine-1-carboxylate (**1l**).* The compound was prepared by the same procedure with **1k** using 2-piperidone instead of 2-pyrrolidinone in Step 1 (309 mg, 57% overall yield, colorless crystals)[12]. Mp: 42−44 °C, purified by chromatography on silica gel (*n*-hexane/EtOAc = 10:1 to 7:1 as the eluent). ^1H NMR ($CDCl_3$, 400 MHz): 6:4 mixture of rotamers, δ 1.35−1.70 (m, 4H), 1.49 (s, 9H), 1.76 (tt, 1H, *J* = 13.8, 4.0 Hz), 1.81−1.95 (m, 1H), 2.40 (t, 1H, *J* = 2.4 Hz), 2.89 (t, 0.6H, *J* = 12.6 Hz), 3.01 (t, 0.4H, *J* = 12.4 Hz), 3.73−4.20 (m, 3H), 5.56 (br s, 0.6H), 5.61 (br s, 0.4H). $^{13}C\{^1H\}$ NMR ($CDCl_3$, 100 MHz): 6:4 mixture of rotamers, δ 18.3, 24.9, 28.3, 30.1, 30.3, 38.0, 39.5, 53.8, 54.8, 73.3, 73.6, 79.9, 80.0, 80.2, 154.4, 155.0. IR (ATR): 3248, 3008, 2979, 2952, 2934, 2916, 2867, 2855, 1682, 1477, 1469, 1449, 1434, 1408, 1375, 1366, 1338, 1316, 1274, 1260, 1234, 1158, 1147, 1137, 1092, 1071, 1056, 1031, 990, 945, 933, 917, 868, 814, 770, 712 cm^{-1}. HRMS (ESI) *m/z*: $[M + Na]^+$ calcd for $C_{13}H_{21}NO_3Na$, 262.1414; found, 262.1411.

*tert-Butyl 2-(Propa-1,2-dien-1-yloxy)piperidine-1-carboxylate (**2l**).* The compound was prepared by the same procedure with **2a** (166 mg, 55% yield, colorless oil) and purified by chromatography on silica gel (*n*-hexane/EtOAc = 15:1 as the eluent). ^1H NMR (acetone-d_6, 400 MHz): 5:5 mixture of rotamers, δ 1.25−1.80 (m, 5H), 1.45 (s, 9H), 1.90 (br, 1H), 2.70−3.10 (br, 1H), 3.84 (br, 1H), 5.32 (br, 2H), 5.64 (br, 1H), 6.61 (br, 1H). $^{13}C\{^1H\}$ NMR (acetone-d_6, 100 MHz): 5:5 mixture of rotamers, δ 19.0, 25.5, 28.5, 29.8, 39.5, 40.6, 80.4, 81.3, 82.4, 88.8, 118.8, 154.8, 155.2, 203.0. IR (ATR): 2976, 2942, 2870, 1697, 1444, 1409, 1391, 1376, 1365, 1339, 1316, 1272, 1254, 1157, 1146, 1090, 1074, 1046, 1017, 989, 958, 930, 868, 814, 769, 723 cm^{-1}. HRMS (ESI) *m/z*: $[M + H]^+$ calcd for $C_{13}H_{22}NO_3$, 240.1594; found, 240.1595.

*Representative Procedure for the Brønsted Acid-Catalyzed Aza-Ferrier Reaction of **2a** into **3a** (Table 1, Entry 6).* Tf_2NH (26 mg 0.09 mmol) was dissolved in CH_2Cl_2 (0.9 mL) at room temperature under an Ar atmosphere. A part of the solution (0.53 mL, 0.053 mmol Tf_2NH) was charged in a flask, and the solution was diluted with CH_2Cl_2 (2.7 mL). This solution was added to a solution of **2a** (160 mg, 0.529 mmol) in CH_2Cl_2 (3.2 mL) at −78 °C under an Ar atmosphere, and the mixture was stirred for 1 h at the same temperature. The resulting mixture was poured into saturated aqueous $NaHCO_3$ and extracted with CH_2Cl_2. The combined extracts were washed with brine, dried over Na_2SO_4, and concentrated by evaporation. Purification of the residue by chromatography on silica gel (*n*-hexane/EtOAc = 3:1 as the eluent) afforded *tert*-butyl (2-formyl-5-phenylpent-1-en-3-yl)(methyl)carba-mate (**3a**) (149 mg, 93% yield) as a colorless oil [in an >1.0 mmol scale (Table 1, entry 7): 354 mg, 93% yield from 1.26 mmol of **2a**]. ^1H NMR ($CDCl_3$, 400 MHz): δ 1.46 (s, 9H), 2.03 (br, 2H), 2.62 (br, 2H), 2.73 (s, 3H), 4.97 (br, 1H), 6.14 (s, 1H), 6.28 (s, 1H), 7.15−7.23 (m, 3H), 7.24−7.33 (m, 2H), 9.54 (s, 1H). $^{13}C\{^1H\}$ NMR ($CDCl_3$, 100 MHz): δ 28.4, 29.1, 30.3, 31.5, 32.6, 52.6, 53.2, 79.7, 80.0, 126.0, 128.3, 128.4, 134.1, 141.2, 141.5, 148.5, 148.8, 155.8, 193.2, 193.4. IR (ATR): 3087, 3061, 3027, 2975, 2930, 2864, 1684, 1626, 1604, 1496, 1478, 1454, 1389, 1364, 1324, 1254, 1154, 1133, 1083, 1045, 1030, 959, 874, 752, 699 cm^{-1}. HRMS (ESI) *m/z*: $[M + H]^+$ calcd for $C_{18}H_{26}NO_3$, 304.1907; found, 304.1901.

*tert-Butyl Benzyl(2-formyl-5-phenylpent-1-en-3-yl)carbamate (**3b**):* 132 mg, 91% yield, colorless oil, purified by chromatography on silica gel (*n*-hexane/EtOAc = 5:1 as the eluent). ^1H NMR (acetone-d_6, 400 MHz): δ 1.47 (br, 9H), 1.92−2.20 (br m, 2H), 2.47 (t, 2H, *J* = 7.4 Hz), 4.39 (br s, 2H), 5.00 (br s, 1H), 6.29 (s, 1H), 6.55 (br s, 1H), 7.02 (br, 2H), 7.10−7.34 (m, 8H), 9.48 (s, 1H). $^{13}C\{^1H\}$ NMR (acetone-d_6, 100 MHz): δ 28.6, 33.59, 33.62, 48.2, 49.2, 53.9, 80.2, 126.6 127.6, 128.0−129.0 (br m), 129.0, 129.1, 129.2, 136.5, 140.9, 142.7, 149.6, 156.1, 156.7, 194.4. IR (ATR): 3087, 3062, 3027, 3004, 2975, 2929, 2863, 2815, 1684, 1604, 1496, 1453, 1430, 1402, 1364, 1331, 1248, 1160, 1117, 1076, 1029, 998, 958, 896, 866, 784, 753, 736, 697 cm^{-1}. HRMS (ESI) *m/z*: $[M + H]^+$ calcd for $C_{24}H_{30}NO_3$, 380.2220; found, 380.2213.

*tert-Butyl Allyl(2-formyl-5-phenylpent-1-en-3-yl)carbamate (**3c**):* 60.0 mg, 86% yield, colorless oil, purified by chromatography on silica gel (*n*-hexane/EtOAc = 5:1 as the eluent). ^1H NMR (acetone-d_6, 400 MHz): δ 1.46 (s, 9H), 1.97−2.22 (m, 2H), 2.63 (t, 2H, *J* = 8.0 Hz), 3.72 (d, 2H, *J* = 5.8 Hz), 4.96 (br, 1H), 5.00−5.07 (br m, 1H), 5.07 (ddt, 1H, *J* = 17.2, 1.6, 1.6 Hz), 5.82 (ddt, 1H, *J* = 17.2, 10.4, 5.8 Hz), 6.34 (s, 1H), 6.55 (d, 1H, *J* = 1.6 Hz), 7.17 (tt, 1H, *J* = 7.0, 1.6 Hz), 7.20−7.30 (m, 4H), 9.56 (s, 1H). $^{13}C\{^1H\}$ NMR (acetone-d_6, 100 MHz): δ 28.7, 33.4, 33.6, 47.8, 53.7, 79.9, 116.5, 126.7, 129.2, 129.3, 136.0, 136.8, 142.8, 149.9, 155.9, 194.4. IR (ATR): 3058, 3063, 3027, 2976, 2929, 2863, 2816, 1685, 1604, 1496, 1477, 1453, 1394, 1364, 1325, 1249, 1161, 1135, 1093, 1030, 992, 957, 920, 863, 770, 752, 699 cm^{-1}. HRMS (ESI) *m/z*: $[M + H]^+$ calcd for $C_{20}H_{28}NO_3$, 330.2064; found, 330.2056.

*tert-Butyl (2-Formylhept-1-en-3-yl)(methyl)carbamate (**3d**):* 73.8 mg, 91% yield, colorless oil, purified by chromatography on silica gel (*n*-hexane/EtOAc = 5:1 to 3:1 as the eluent). ^1H NMR ($CDCl_3$, 400 MHz): δ 0.91 (t, 3H, *J* = 7.0 Hz), 1.22−1.55 (m, 4H), 1.47 (s, 9H), 1.70 (br, 2H), 2.68 (s, 3H), 4.93 (br, 1H), 6.14 (s, 1H), 6.27 (d, 1H, *J* = 1.2 Hz), 9.55 (d, 1H, *J* = 0.4 Hz). $^{13}C\{^1H\}$ NMR ($CDCl_3$, 100 MHz): δ 13.9, 22.2, 28.2, 28.3, 29.1, 52.6, 79.6, 133.7, 149.2, 155.8, 193.2. IR (ATR): 2959, 2930, 2863, 1685, 1626, 1479, 1455, 1427, 1389, 1364, 1334, 1254, 1178, 1144, 1113, 1055, 1021, 961, 943, 904, 875, 835, 770 cm^{-1}. HRMS (ESI) *m/z*: $[M + H]^+$ calcd for $C_{14}H_{26}NO_3$, 256.1907; found, 256.1903.

*tert-Butyl (3-Formylbut-3-en-2-yl)(methyl)carbamate (**3e**):* 48.9 mg, 89% yield, colorless oil, purified by chromatography on silica gel (*n*-hexane/EtOAc = 4:1 to 2:1 as the eluent). In an >1.0 mmol scale: 279 mg, 88% yield from 1.49 mmol of **2e**. ^1H NMR ($CDCl_3$, 400 MHz): δ 1.34 (d, 3H, *J* = 7.2 Hz), 1.46 (s, 9H), 2.72 (s, 3H), 5.04 (br, 1H), 6.13 (s, 1H), 6.25 (d, 1H, *J* = 1.6 Hz), 9.57 (s, 1H). $^{13}C\{^1H\}$ NMR ($CDCl_3$, 100 MHz): δ 15.9, 28.3, 29.3, 48.8, 79.6, 133.2, 150.1, 155.4, 193.1. IR (ATR): 2976, 2933, 1683, 1479, 1441, 1389, 1364, 1348, 1323, 1254, 1143, 1092, 1081, 1045, 1031, 983, 947, 871, 842, 792, 773 cm^{-1}. HRMS (ESI) *m/z*: $[M + H]^+$ calcd for $C_{11}H_{20}NO_3$, 214.1438; found, 214.1438.

*tert-Butyl (3-Formyl-1-phenylbut-3-en-2-yl)(methyl)carbamate (**3f**):* 60.1 mg, 75% yield, colorless oil, purified by chromatography on silica gel (*n*-hexane/EtOAc = 3:1 as the eluent). ^1H NMR ($CDCl_3$, 400 MHz): 6:4 mixture of rotamers, δ 1.31 (s, 9H), 2.67 (s, 3H), 2.99 (br, 1.2H), 3.11 (br, 0.8H), 5.07 (br, 0.4H), 5.21 (br, 0.6H), 6.20 (s, 1H), 6.39 (br, 1H), 7.13−7.31 (m, 5H), 9.55 (s, 1H). $^{13}C\{^1H\}$ NMR ($CDCl_3$, 100 MHz): 6:4 mixture of rotamers, δ 28.1, 29.6, 32.6, 35.4, 35.6, 54.4, 55.6, 79.5, 79.6, 126.4, 128.3, 128.9, 134.4, 135.2, 137.9, 148.0, 148.5, 155.3, 193.0, 193.7. IR (ATR): 3028, 3003, 2975, 2929, 1683, 1604, 1496, 1478, 1455, 1429, 1389, 1364, 1338, 1251, 1168, 1136, 1079, 1052, 1030, 954, 914, 872, 824, 767, 747, 698 cm^{-1}. HRMS (ESI) *m/z*: $[M + H]^+$ calcd for $C_{17}H_{24}NO_3$, 290.1751; found, 290.1744.

*tert-Butyl (2-Formyl-1-phenylallyl)(methyl)carbamate (**3g**):* 86.1 mg, 69% yield, white solid. Mp: 55−58 °C, purified by chromatography on silica gel (*n*-hexane/EtOAc = 3:1 as the eluent). ^1H NMR ($CDCl_3$, 400 MHz): δ 1.45 (s, 9H), 2.66 (s, 3H), 6.21 (br, 1H), 6.23 (s, 1H), 6.33 (d, 1H, *J* = 0.8 Hz), 7.12−7.17 (m, 2H), 7.27 (tt, 1H, *J* = 7.2, 1.2 Hz), 7.30−7.36 (m, 2H), 9.67 (s, 1H). $^{13}C\{^1H\}$ NMR ($CDCl_3$, 100 MHz): δ 28.3, 31.6, 57.7, 80.1, 127.5, 128.1, 128.5, 135.2, 137.4, 149.0, 155.6, 192.5. IR (ATR): 2975, 2931, 1685, 1478, 1452, 1440, 1384, 1365, 1350, 1321, 1272, 1253, 1174, 1138, 1079, 1045, 1028, 1003, 965, 922, 885, 865, 844, 769, 738, 698 cm^{-1}. HRMS (ESI) *m/z*: $[M + H]^+$ calcd for $C_{16}H_{22}NO_3$, 276.1594; found, 276.1589.

*tert-Butyl (2-Formyl-4-methylpent-1-en-3-yl)(methyl)carbamate (**3h**):* 128 mg, 89% yield, colorless oil, purified by chromatography on silica gel (*n*-hexane/EtOAc = 5:1 to 3:1 as the eluent). ^1H NMR (acetone-d_6, 400 MHz): δ 0.81 (d, 3H, *J* = 5.6 Hz), 0.92 (s, 3H), 1.44 (s, 9H), 2.33 (br, 1H), 2.63 (s, 3H), 4.62 (d, 1H, *J* = 11.2 Hz), 6.39 (s, 1H), 6.54 (s, 1H), 9.57 (s, 1H). $^{13}C\{^1H\}$ NMR ($CDCl_3$, 100

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MHz): mixture of rotamers, δ 19.2, 20.0, 26.6, 27.1, 28.3, 29.3, 30.9, 58.5, 59.7, 79.3, 79.7, 135.1, 136.2, 147.3, 155.7, 193.6, 194.1. IR (ATR): 2969, 2932, 2875, 1685, 1470, 1454, 1441, 1388, 1365, 1334, 1297, 1254, 1146, 1115, 1045, 1027, 978, 947, 890, 874, 833, 768 cm^{-1}. HRMS (ESI) *m/z*: $[M + H]^+$ calcd for $C_{13}H_{24}NO_3$, 242.1751; found, 242.1746.

*tert-Butyl (2-Formylallyl)(methyl)carbamate (**3i**):* 81.4 mg, 81% yield, colorless oil, purified by chromatography on silica gel (*n*-hexane/EtOAc = 7:1 to 5:1 as the eluent). In an >1.0 mmol scale: 196 mg, 86% yield from 1.14 mmol of **2i**. ^1H NMR ($CDCl_3$, 400 MHz): 6:4 mixture of rotamers, δ 1.43 (s, 5.4H), 1.47 (s, 3.6H), 2.88 (s, 3H), 4.00−4.11 (br, 2H), 6.15 (t, 1H, *J* = 1.0 Hz), 6.23 (br, 0.6H), 6.26 (br, 0.4H), 9.65 (s, 1H). $^{13}C\{^1H\}$ NMR ($CDCl_3$, 100 MHz): 6:4 mixture of rotamers, δ 28.2, 34.6, 35.0, 46.1, 46.7, 79.7, 133.2, 133.9, 145.3, 145.8, 155.6, 155.7, 193.7. IR (ATR): 2976, 2931, 1683, 1480, 1453, 1418, 1389, 1365, 1323, 1301, 1264, 1243, 1228, 1174, 1143, 1051, 1033, 955, 881, 853, 772, 685 cm^{-1}. HRMS (ESI) *m/z*: $[M + H]^+$ calcd for $C_{10}H_{18}NO_3$, 200.1281; found, 200.1280.

*tert-Butyl ((5R)-2-Formyl-5,9-dimethyldeca-1,8-dien-3-yl)-(methyl)carbamate (**3j**):* 3.0 mg, 2% yield, colorless oil, purified by chromatography on silica gel (*n*-hexane/EtOAc = 5:1 to 3:1 as the eluent). ^1H NMR ($CDCl_3$, 400 MHz): mixture of diastereomers and rotamers, δ 0.92 (d, 3H, *J* = 6.8 Hz), 0.97 (d, 3H, *J* = 6.0 Hz), 1.05−1.85 (m, 5H), 1.47 (s, 9H), 1.597 (s, 1.5H), 1.603 (s, 1.5H), 1.68 (d, 3H, *J* = 0.8 Hz), 1.85−2.12 (m, 2H), 2.666 (s, 1.5H), 2.671 (s, 1.5H), 5.04−5.12 (m, 2H), 6.12 (d, 0.5H, *J* = 1.0 Hz), 6.13 (s, 0.5H), 6.24 (d, 0.5H, *J* = 1.0 Hz), 6.26 (d, 0.5H, *J* = 1.0 Hz), 9.540 (s, 0.5H), 9.543 (s, 0.5H). $^{13}C\{^1H\}$ NMR: not performed due to a small amount. IR (ATR): 2965, 2924, 2871, 2854, 1687, 1478, 1453, 1437, 1389, 1364, 1325, 1283, 1253, 1145, 1057, 955, 914, 873, 829, 810, 771, 742, 701 cm^{-1}. HRMS (ESI) *m/z*: $[M + H]^+$ calcd for $C_{19}H_{34}NO_3$, 324.2533; found, 324.2531.

*tert-Butyl 2-(3-Oxoprop-1-en-2-yl)pyrrolidine-1-carboxylate (**3k**):* 178 mg, 94% yield, colorless oil, purified by chromatography on silica gel (*n*-hexane/EtOAc = 5:1 to 3:1 as the eluent). ^1H NMR ($CDCl_3$, 400 MHz): 6:4 mixture of rotamers, δ 1.37 (s, 5.4H), 1.46 (s, 3.6H), 1.60−1.92 (m, 3H), 2.09−2.25 (m, 1H), 3.35−3.58 (m, 2H), 4.68 (d, 0.6H, *J* = 8.0 Hz), 4.72 (d, 0.4H, *J* = 7.6 Hz), 6.05 (s, 1H), 6.15 (s, 0.4H), 6.17 (s, 0.6H), 9.62 (s, 1H). $^{13}C\{^1H\}$ NMR ($CDCl_3$, 100 MHz): 6:4 mixture of rotamers, δ 22.5, 23.3, 28.3, 28.4, 31.0, 32.0, 46.4, 46.8, 55.1, 55.4, 79.4, 132.3, 132.5, 150.0, 151.5, 154.1, 193.4, 193.5. IR (ATR): 2975, 2932, 2878, 1682, 1477, 1453, 1388, 1363, 1298, 1257, 1159, 1117, 1089, 1031, 947, 914, 877, 853, 770, 738 cm^{-1}. HRMS (ESI) *m/z*: $[M + H]^+$ calcd for $C_{12}H_{20}NO_3$, 226.1438; found, 226.1435.

*tert-Butyl 2-(3-Oxoprop-1-en-2-yl)piperidine-1-carboxylate (**3l**):* 154 mg, 93% yield, colorless oil, purified by chromatography on silica gel (*n*-hexane/EtOAc = 5:1 to 3:1 as the eluent). ^1H NMR ($CDCl_3$, 400 MHz): δ 1.26 (ddddd, 1H, *J* = 12.6, 12.6, 12.6, 3.8, 3.8 Hz), 1.35−1.62 (m, 2H), 1.42 (s, 9H), 1.62−1.80 (m, 2H), 2.01 (br d, 1H, *J* = 14.0 Hz), 2.94 (ddd, 1H, *J* = 13.2, 12.6, 3.8 Hz), 4.07 (br dd, 1H, *J* = 12.6, 3.4 Hz), 5.08 (br d, 1H, *J* = 5.6 Hz), 6.13 (d, 1H, *J* = 1.6 Hz), 6.20 (d, 1H, *J* = 1.6 Hz), 9.57 (s, 1H). $^{13}C\{^1H\}$ NMR ($CDCl_3$, 100 MHz): δ 19.1, 24.9, 26.9, 28.3, 40.8, 50.1, 79.7, 133.9, 149.3, 155.2, 193.7. IR (ATR): 2975, 2934, 2862, 1683, 1473, 1454, 1391, 1363, 1270, 1253, 1159, 1120, 1091, 1074, 1042, 996, 948, 930, 918, 873, 843, 821, 774, 735, 697 cm^{-1}. HRMS (ESI) *m/z*: $[M + H]^+$ calcd for $C_{13}H_{22}NO_3$, 240.1594; found, 240.1591.

*Wittig Reaction of **3e** to Produce tert-Butyl Methyl(3-methylenepent-4-en-2-yl)carbamate (**4**) (Scheme 4a).* A suspension of methyltriphenylphosphonium iodide (404 mg, 0.992 mmol) in Et_2O (5 mL) was treated with *t*BuOK (112 mg, 1.00 mmol) at room temperature under an Ar atmosphere and stirred for 15 min. The mixture was cooled at 0 °C and a solution of **3e** (106 mg, 0.497 mmol) in Et_2O (2.5 mL) was added to the mixture. After stirring for 18 h at room temperature, the resulting mixture was quenched with saturated aqueous NH_4Cl and extracted with EtOAc. The combined extracts were washed with brine, dried over Na_2SO_4, and concentrated by evaporation. The residue was purified by chromatography on silica gel (*n*-hexane/EtOAc = 10:1 as the eluent) to obtain **4** (82.3 mg, 78% yield) as a colorless oil. ^1H NMR ($CDCl_3$, 400 MHz): 6:4 mixture of rotamers, δ 1.25 (d, 3H, *J* = 6.8 Hz), 1.48 (s, 9H), 2.54 (s, 3H), 4.93 (br, 0.4H), 5.00−5.21 (br, 0.6H), 5.06 (d, 1H, *J* = 11.2 Hz), 5.12 (d, 1H, *J* = 0.8 Hz), 5.26 (s, 1H), 5.29−5.48 (br, 1H), 6.29 (dd, 1H, *J* = 17.6, 11.2 Hz). $^{13}C\{^1H\}$ NMR ($CDCl_3$, 100 MHz): 6:4 mixture of rotamers, δ 15.3, 15.6, 27.5, 28.4, 49.0, 50.3, 79.2, 114.3, 114.6, 116.3, 137.1, 146.1, 155.5. IR (ATR): 3094, 3006, 2976, 2933, 1684, 1596, 1478, 1454, 1441, 1424, 1389, 1365, 1320, 1255, 1181, 1141, 1081, 1045, 1029, 987, 906, 865, 833, 771, 733 cm^{-1}. HRMS (ESI) *m/z*: $[M + H]^+$ calcd for $C_{12}H_{22}NO_2$, 212.1645; found, 212.1642.

*Diels−Alder Reaction of **4** to Produce tert-Butyl (1-((rel-4aR,9aS)-9,10-Dioxo-1,4,4a,9,9a,10-hexahydroanthracen-2-yl)-ethyl)(methyl)carbamate (**5**) (Scheme 4a).* A solution of **4** (61.7 mg, 0.292 mmol) and 1,4-naphthoquinone (46 mg, 0.29 mmol) in toluene (3 mL) was refluxed for 1 day under an Ar atmosphere. The resulting solution was cooled at room temperature and concentrated by evaporation. The residue was purified by chromatography on silica gel (*n*-hexane/EtOAc = 3:1 to 2:1 as the eluent) to afford **5** (91.0 mg, 85% yield) as a pale brown amorphous solid. The compound **5** would be a 1:1 mixture of diastereomers. ^1H NMR ($CDCl_3$, 400 MHz): mixture of diastereomers and rotamers, δ 1.19 (d, 1.5H, *J* = 6.8 Hz), 1.20 (d, 1.5H, *J* = 6.8 Hz), 1.44 (s, 4.5H), 1.45 (s, 4.5H), 2.00−2.20 (br m, 1H), 2.20−2.90 (br m, 3H), 2.54 (s, 1.5H), 2.58 (s, 1.5H), 3.25−3.55 (m, 2H), 4.25−4.95 (br, 1H), 5.57−5.66 (m, 1H), 7.71−7.80 (m, 2H), 7.99−8.08 (m, 2H). $^{13}C\{^1H\}$ NMR (acetone-d_6, 100 MHz): mixture of diastereomers and rotamers, δ 15.2, 25.5, 25.6, 26.7, 27.8, 28.0, 28.65, 28.67, 47.0, 47.2, 47.3, 47.5, 53.9, 55.2, 79.3, 79.4, 121.0, 121.2, 127.3, 127.4, 134.88, 134.90, 135.0, 135.1, 135.2, 136.5, 136.6, 156.3, 198.07, 198.13, 198.3, 198.4. IR (ATR): 3065, 2973, 2929, 2886, 2838, 1683, 1593, 1478, 1453, 1438, 1423, 1389, 1364, 1319, 1282, 1248, 1141, 1078, 1054, 1024, 1011, 973, 954, 937, 872, 863, 824, 815, 794, 771, 737, 719, 690 cm^{-1}. HRMS (ESI) *m/z*: $[M + H]^+$ calcd for $C_{22}H_{28}NO_4$, 370.2013; found, 370.2011.

*1,2-Reduction of **3l** to Produce tert-Butyl 2-(3-Hydroxyprop-1-en-2-yl)piperidine-1-carboxylate (**6**) (Scheme 4b).* A solution of **3l** (45.5 mg, 0.190 mmol) and $CeCl_3 \cdot 7H_2O$ (85 mg, 0.23 mmol) in MeOH (0.95 mL) was treated with $NaBH_4$ (12 mg, 0.32 mmol) at 0 °C and stirred for 30 min under an Ar atmosphere. The resulting mixture was quenched with saturated aqueous NH_4Cl and extracted with EtOAc. The combined extracts were washed with brine, dried over Na_2SO_4, and concentrated by evaporation. The residue was purified by chromatography on silica gel (*n*-hexane/EtOAc = 2:1 as the eluent) to obtain **6** (40.4 mg, 88% yield) as a colorless oil. ^1H NMR ($CDCl_3$, 400 MHz): δ 1.37−1.53 (m, 1H), 1.47 (s, 9H), 1.53−1.66 (m, 3H), 1.66−1.78 (m, 1H), 2.06 (br d, 1H, *J* = 13.6 Hz), 2.64 (ddd, 1H, *J* = 12.9, 12.9, 2.6 Hz), 2.90−3.80 (br, 1H), 3.87−4.06 (m, 3H), 4.91 (br, 1H), 5.01 (d, 1H, *J* = 1.0 Hz), 5.30 (d, 1H, *J* = 1.0 Hz). $^{13}C\{^1H\}$ NMR ($CDCl_3$, 100 MHz): δ 19.6, 25.5, 25.7, 28.4, 40.4, 50.4, 64.8, 80.1, 113.2, 146.2, 155.9. IR (ATR): 3428, 2974, 2935, 2864, 1687, 1665, 1475, 1453, 1410, 1364, 1337, 1325, 1311, 1276, 1252, 1158, 1135, 1122, 1057, 1030, 994, 903, 870, 813, 769 cm^{-1}. HRMS (ESI) *m/z*: $[M + Na]^+$ calcd for $C_{13}H_{23}NO_3Na$, 264.1570; found, 264.1563.

*Cyclization of **6** to Produce 4-Methylenehexahydropyrido[1,2-c][1,3]oxazin-1(3H)-one (**7**) (Scheme 4b).* A solution of **6** (40.2 mg, 0.167 mmol) in THF (0.85 mL) was treated with NaH (60 wt % in oil, 8 mg, 0.2 mmol) at 0 °C. The mixture was stirred for 1 h at 0 °C and for 2 h at room temperature. The resulting mixture was quenched with saturated aqueous NH_4Cl at 0 °C and extracted with EtOAc. The combined extracts were washed with brine, dried over Na_2SO_4, and concentrated by evaporation. Purification of the residue by chromatography on silica gel (*n*-hexane/EtOAc = 1:2 as the eluent) gave **7** (20.6 mg, 74% yield) as a colorless oil. ^1H NMR ($CDCl_3$, 400 MHz): δ 1.44−1.63 (m, 3H), 1.65−1.75 (m, 1H), 1.79−1.89 (m, 1H), 1.89−2.00 (m, 1H), 2.63−2.76 (m, 1H), 3.83 (d, 1H, *J* = 10.4 Hz), 4.34 (dddd, 1H, *J* = 13.6, 4.0, 2.0, 2.0 Hz), 4.48 (dd, 1H, *J* = 12.0, 0.4 Hz), 4.63 (ddd, 1H, *J* = 12.0, 0.8, 0.4 Hz), 5.12 (d, 1H, *J* = 1.4 Hz), 5.15 (ddd, 1H, *J* = 1.4, 0.8, 0.8 Hz). $^{13}C\{^1H\}$ NMR ($CDCl_3$, 100 MHz): δ 24.0, 24.8, 33.6, 45.7, 59.1, 68.3, 112.1, 139.7, 154.5. IR (ATR): 2991, 2936, 2856, 1683, 1466, 1426, 1362, 1349, 1336, 1316,

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1306, 1276, 1244, 1223, 1166, 1141, 1119, 1097, 1034, 988, 914, 855, 840, 761 cm^{-1}. HRMS (ESI) *m/z*: $[M + H]^+$ calcd for $C_9H_{14}NO_2$, 168.1019; found, 168.1014.

*Michael Addition of Diethyl Malonate with **3i** occurred to Provide Diethyl 2-(3′-((tert-Butoxycarbonyl)(methyl)amino)-2′-formylpropyl)malonate (**8**) (Scheme 4c).* A solution of diethyl malonate (47 μL, 0.31 mmol) in EtOH (1.5 mL) was treated with EtONa (2 mg, 0.03 mmol) at room temperature under an Ar atmosphere and stirred for 5 min. A solution of **3i** (61.6 mg, 0.309 mmol) in EtOH (1.5 mL) was added, and the mixture was stirred for 6 h at room temperature. The resulting mixture was quenched with saturated aqueous NH_4Cl and extracted with EtOAc. The combined extracts were washed with brine, dried over Na_2SO_4, and concentrated by evaporation. Purification of the residue by chromatography on silica gel (*n*-hexane/EtOAc = 4:1 to 2:1 as the eluent) gave **8** (87.6 mg, 79% yield) as a colorless oil. ^1H NMR ($CDCl_3$, 400 MHz): δ 1.27 (t, 6H, *J* = 7.2 Hz), 1.45 (s, 9H), 1.97 (ddd, 1H, *J* = 13.9, 8.4, 4.8 Hz), 2.28 (ddd, 1H, *J* = 13.9, 7.2, 7.2 Hz), 2.68 (br, 1H), 2.87 (s, 3H), 3.28−3.41 (br, 1H), 3.41−3.80 (br m, 2H), 4.10−4.30 (m, 4H), 9.66 (s, 1H). $^{13}C\{^1H\}$ NMR ($CDCl_3$, 100 MHz): 5:5 mixture of rotamers, δ 13.9, 25.5, 28.1, 34.9, 35.2, 48.2, 48.8, 49.1, 49.3, 61.5, 79.9, 80.4, 155.0, 155.7, 168.7, 201.8, 202.0. IR (ATR): 2979, 2935, 2875, 1727, 1688, 1480, 1455, 1392, 1366, 1300, 1226, 1146, 1096, 1025, 932, 865, 773, 733 cm^{-1}. HRMS (ESI) *m/z*: $[M + H]^+$ calcd for $C_{17}H_{30}NO_7$, 360.2017; found, 360.2014.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.joc.0c01047.

Synthetic schemes of **1** and **2**; copies of NMR (^1H, ^{13}C) spectra for **1**−**8** (PDF)

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Notes

The authors declare no competing financial interest.

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(9) Substrate **1a** was prepared from 3-phenylpropionaldehyde in 3 steps (80% overall yield). Substrates **1b**−**j** were also prepared from the corresponding aldehydes (22−78% overall yield). Substrates **1k** and **1l** were prepared from 2-pyrrolidinone or 2-piperidone in 3 steps, respectively (48−57% overall yield).

(10) For representative examples of isomerization of *O,O*-propargyl ethers into *O,O*-allenyl acetals, see: (a) Takeda, K.; Nakajima, A.; Takeda, M.; Yoshii, E. [3 + 4] Annulation using a [β-(Trimethylsilyl) acryloyl]silane and the Lithium Enolate of an α,β-Unsaturated Methyl Ketone: (1*R*,6*S*,7*S*)-4-(*tert*-Butyldimethylsiloxy)-6-(trimethylsilyl)-bicyclo[5.4.0]undec-4-en-2-one. *Org. Synth.* **1999**, *76*, 199−206. (b) Mereyala, H. B.; Gurrala, S. R.; Mohan, S. K. Study of Metal and Acid Catalysed Deprotection of Propargyl Ethers of Alcohols via their Allenyl Ethers. *Tetrahedron* **1999**, *55*, 11331−11342. (c) Bee, C.; Tius, M. A. Convergent Cyclopentannelation Process. *Org. Lett.* **2003**, *5*, 1681−1684. (d) Deagostino, A.; Prandi, C.; Toppino, A.; Venturello, P. Palladium-Catalysed Heck Reaction on 1,2-Dien-1-ols: A Stereoselective Synthesis of α-Arylated α,β-Unsaturated Aldehydes. *Tetrahedron* **2008**, *64*, 10344−10349. (e) Boi, T.; Deagostino, A.; Prandi, C.; Tabasso, S.; Toppino, A.; Venturello, P. Heck Reaction on Protected 3-Alkyl-1,2-dien-1-ols: An Approach to Substituted 3-Alkenylindoles, 2-Alkoxy-3-alkylidene-2,3-dihydrobenzofuranes and -indolidines. *Org. Biomol. Chem.* **2010**, *8*, 2020−2027. (f) Brasholz, M.; Dugovic, B.; Reissig, H. 3-Alkoxy-2,5-dihydrofurans by Gold-Catalyzed Allenyl Cyclizations and Their Transformation into 1,4-Dicarbonyl Compounds, Cyclopentenones, and Butenolides. *Synthesis* **2010**, *2010*, 3855−3864. (g) Lechel, T.; Gerhard, M.; Trawny, D.; Brusilowskij, B.; Schefzig, L.; Zimmer, R.; Rabe, J. P.; Lentz, D.; Schalley, C. A.; Reissig, H. Synthesis of 5-Acetyloxazoles and 1,2-Diketones from β-Alkoxy-β-ketoenamides and Their Sub-

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